

Embedding Method - Feature Extraction.

Embedding training is a feature extraction method, we can say it is a deep learning method where we train the vector of each word per batch and get the optimum value through backpropagation.

How are embeddings updated?

Standard flow :

- Input Sentence -> Embedding -> CNN -> Dense -> Softmax. The above is a standard flow of embedding + cnn deep learning model.
- Creating an embedding matrix $E \in \mathbb{R}^{V \times d}$
- V = vocabulary size
- d = embedding dimension

Compute Loss:

- if we use crossentropy = $L = \text{crossentropy}(y, \hat{y})$

Compute Gradients wrt loss and vector update:

- $\frac{\partial E[i]}{\partial L}$
- gradient: $g_i = \frac{\partial L}{\partial E[i]}$
- If optimizer is SGD: $E[i] := E[i] - \eta \cdot g_i$
- If optimizer is ADAM: $E[i] := E[i] - \eta \cdot \frac{m_i}{\sqrt{v_i} + \epsilon}$

Model Training:

Using this vectorizing method we will train Deep Learning Models:

- CNN - We will use a 1D Convolutional Model with kernel of dimension 3x1
- MLP - Using a standard MLP Architecture with 256, 128 neurons in input and hidden layer.

Embedding Method – Feature Extraction

Before training deep learning models, raw text must be converted into numeric form. We use the Tokenizer + Embedding method to achieve this.

$\text{Input} \rightarrow \text{Tokenizer} \rightarrow \text{Embedding} \rightarrow \text{CNN / MLP} \rightarrow \text{Softmax}$

Tokenizer

The Keras Tokenizer preprocesses and converts text into sequences of integers by:

- lowercasing text
- removing punctuation
- splitting into words (tokens)
- building a vocabulary of the top V most frequent words
- mapping each word to an integer ID
- padding sequences to a fixed length

Each sentence becomes: $\text{sentence} \rightarrow [w_1, w_2, \dots, w_T]$ Embedding Layer

An embedding matrix is initialized:

Where:

- V = vocabulary size
- d = embedding dimension

Each integer token i retrieves vector $E[i]$. These vectors are learned during training using backpropagation.

Updating Embeddings

Given loss L (crossentropy):

$$L = -\sum_c y_c \log(\hat{y}_c)$$

$$\text{SGD update: } E[i] := E[i] - \eta \cdot g_i$$

$$\text{Adam update: } E[i] := E[i] - \eta \cdot \frac{m_i}{\sqrt{v_i} + \epsilon}$$

Only vectors corresponding to words in the batch are updated.

Model Training

Using this vectorized representation, we train:

CNN Model

Embedding $\rightarrow$ Conv1D(kernel=3) $\rightarrow$ GlobalMaxPool $\rightarrow$ Dense $\rightarrow$ Softmax

MLP Model

Embedding → Dense(256) → Dense(128) → Dense(output)

Both models learn classification patterns using the embedding vectors as features.

```
In [1]: import pandas as pd
import numpy as np
import random
import re
from datasets import load_dataset
import time
from sklearn.metrics import accuracy_score, f1_score, classification_report
```

C:\Users\moham\anaconda3\envs\cuda-env\lib\site-packages\tqdm\auto.py:21: TqdmWarning: IProgress not found. Please update jupyter and ipywidgets. See https://ipywidget.s.readthedocs.io/en/stable/user_install.html
 from .autonotebook import tqdm as notebook_tqdm

```
In [2]: #Loading dataset

dataset = load_dataset('ag_news')
train_df = dataset['train'].to_pandas()
test_df = dataset['test'].to_pandas()

X_train = train_df['text']
y_train = train_df['label']
X_test = test_df['text']
y_test = test_df['label']
```

```
In [3]: X_train
```

```
Out[3]: 0      Wall St. Bears Claw Back Into the Black (Reute...
1      Carlyle Looks Toward Commercial Aerospace (Reu...
2      Oil and Economy Cloud Stocks' Outlook (Reuters...
3      Iraq Halts Oil Exports from Main Southern Pipe...
4      Oil prices soar to all-time record, posing new...
      ...
119995  Pakistan's Musharraf Says Won't Quit as Army C...
119996  Renteria signing a top-shelf deal Red Sox gene...
119997  Saban not going to Dolphins yet The Miami Dolp...
119998  Today's NFL games PITTSBURGH at NY GIANTS Time...
119999  Nets get Carter from Raptors INDIANAPOLIS -- A...
Name: text, Length: 120000, dtype: object
```

```
In [4]: from tensorflow.keras.preprocessing.text import Tokenizer
from tensorflow.keras.preprocessing.sequence import pad_sequences

vocab_size = 65000          # enough for AG News
max_len = 200              # sentence length
oov_token = "<OOV>"

tokenizer = Tokenizer(num_words=vocab_size, oov_token=oov_token)
tokenizer.fit_on_texts(X_train)

# convert text → integer sequences
X_train_seq = tokenizer.texts_to_sequences(X_train)
```

```
X_test_seq = tokenizer.texts_to_sequences(X_test)

# pad to fixed length
X_train_pad = pad_sequences(X_train_seq, maxlen=max_len, padding='post')
X_test_pad = pad_sequences(X_test_seq, maxlen=max_len, padding='post')
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In [5]: X_train_seq
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In [6]: X_train_pad

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Out[6]: array([[ 443,  442, 1682, ...,  0,  0,  0],
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```
In [7]: from tensorflow.keras.utils import to_categorical
```

```
num_classes = 4 # AG News has 4 categories
y_train_oh = to_categorical(y_train, num_classes)
y_test_oh = to_categorical(y_test, num_classes)
```

```
In [8]: import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Embedding, Conv1D, GlobalMaxPooling1D, Dense, Dropout

embedding_dim = 128
filters = 128
kernel_size = 3

cnn_model = Sequential([
    Embedding(vocab_size, embedding_dim, input_length=max_len, name="embedding"),

    Conv1D(filters=filters, kernel_size=kernel_size, activation='relu'),
    GlobalMaxPooling1D(),

    Dense(64, activation='relu'),
    Dropout(0.5),

    Dense(num_classes, activation='softmax')
])

cnn_model.compile(
    optimizer='adam',
    loss='categorical_crossentropy',
    metrics=['accuracy']
)

cnn_model.summary()
```

Model: "sequential"

Layer (type)	Output Shape	Param #
=====		
embedding (Embedding)	(None, 200, 128)	8320000
conv1d (Conv1D)	(None, 198, 128)	49280
global_max_pooling1d (GlobalMaxPooling1D)	(None, 128)	0
dense (Dense)	(None, 64)	8256
dropout (Dropout)	(None, 64)	0
dense_1 (Dense)	(None, 4)	260
=====		
Total params: 8,377,796		
Trainable params: 8,377,796		
Non-trainable params: 0		

```
In [9]: %%time
cpu_start_time = time.process_time()
start_time = time.time()

history = cnn_model.fit(
    X_train_pad, y_train_oh,
    validation_split=0.1,
    epochs=5,
    batch_size=1000
)

cpu_end_time = time.process_time()
end_time = time.time()
cpu_time_train = cpu_end_time - cpu_start_time
cell_time_train = end_time - start_time
print(cpu_time_train)
print(cell_time_train)
```

```

Epoch 1/5
108/108 [=====] - 25s 223ms/step - loss: 0.7658 - accuracy:
0.7421 - val_loss: 0.2768 - val_accuracy: 0.9058
Epoch 2/5
108/108 [=====] - 33s 304ms/step - loss: 0.2557 - accuracy:
0.9230 - val_loss: 0.2421 - val_accuracy: 0.9160
Epoch 3/5
108/108 [=====] - 46s 421ms/step - loss: 0.1719 - accuracy:
0.9483 - val_loss: 0.2507 - val_accuracy: 0.9139
Epoch 4/5
108/108 [=====] - 30s 283ms/step - loss: 0.1157 - accuracy:
0.9657 - val_loss: 0.2720 - val_accuracy: 0.9122
Epoch 5/5
108/108 [=====] - 30s 277ms/step - loss: 0.0783 - accuracy:
0.9772 - val_loss: 0.3047 - val_accuracy: 0.9062
1685.796875
163.31832146644592
CPU times: total: 28min 5s
Wall time: 2min 43s

```

```

In [10]: %%time
cpu_start_time = time.process_time()
start_time = time.time()

y_pred_probs = cnn_model.predict(X_test_pad)

cpu_end_time = time.process_time()
end_time = time.time()
cpu_time_inf = cpu_end_time - cpu_start_time
cell_time_inf = end_time - start_time
print(cpu_time_inf)
print(cell_time_inf)

```

```

238/238 [=====] - 1s 6ms/step
7.875
1.5171289443969727
CPU times: total: 7.88 s
Wall time: 1.52 s

```

```

In [11]: y_pred_test = np.argmax(y_pred_probs, axis=1)

y_train_probs = cnn_model.predict(X_train_pad)
y_train_pred = np.argmax(y_train_probs, axis=1)

```

```

3750/3750 [=====] - 22s 6ms/step

```

```

In [12]: y_train_oh

```

```

Out[12]: array([[0., 0., 1., 0.],
               [0., 0., 1., 0.],
               [0., 0., 1., 0.],
               ...,
               [0., 1., 0., 0.],
               [0., 1., 0., 0.],
               [0., 1., 0., 0.]], dtype=float32)

```

```

In [13]: y_train_pred

```

Out[13]: array([2, 2, 2, ..., 1, 1, 1], dtype=int64)

```
In [14]: print("Training Accuracy")
train_accuracy = accuracy_score(y_train, y_train_pred)
train_f1 = f1_score(y_train, y_train_pred, average='weighted', zero_division=0)
train_report = classification_report(y_train, y_train_pred)
print('\n Training Accuracy',train_accuracy)
print('\n Training F1 Score',train_f1)
print('\n Training Classification Report',train_report)
```

Training Accuracy

Training Accuracy 0.982125

Training F1 Score 0.9821182280113712

Training Classification Report	precision	recall	f1-score	support
0	0.99	0.98	0.98	30000
1	0.99	1.00	0.99	30000
2	0.98	0.97	0.97	30000
3	0.97	0.99	0.98	30000
accuracy			0.98	120000
macro avg	0.98	0.98	0.98	120000
weighted avg	0.98	0.98	0.98	120000

```
In [15]: print("Test Accuracy")
test_accuracy = accuracy_score(y_test, y_pred_test)
test_f1 = f1_score(y_test, y_pred_test, average='weighted', zero_division=0)
test_report = classification_report(y_test,y_pred_test)
print('\n Test Accuracy',test_accuracy)
print('\n Test F1 Score',test_f1)
print('\n Test Classification Report',test_report)
```

Test Accuracy

Test Accuracy 0.9121052631578948

Test F1 Score 0.9120221391223258

Test Classification Report	precision	recall	f1-score	support
0	0.92	0.91	0.91	1900
1	0.96	0.97	0.97	1900
2	0.89	0.87	0.88	1900
3	0.88	0.90	0.89	1900
accuracy			0.91	7600
macro avg	0.91	0.91	0.91	7600
weighted avg	0.91	0.91	0.91	7600

In []:

Per each sample ,the FLOPs carried can be calculated by the formula below. This is entirely dependent on the architecture. $\text{FLOPs} = 2 \times (\text{sequence length}) \times (\text{kernel size}) \times (\text{embedding dim}) \times (\text{filters})$

```
In [16]: flops = 2 * 200 * 3 * 128 * 128
print("The FLOPs carried out per sample is: ", flops)
```

The FLOPs carried out per sample is: 19660800

```
In [17]: results = {}
results[('Matrix Embedding', 'CNN Model')] = {
    'Train':
        {'accuracy': train_accuracy,
         'f1_score': train_f1,
         'report': train_report,
         'training_time': cpu_time_train},
    'Test':
        {'accuracy': test_accuracy,
         'f1_score': test_f1,
         'report': test_report,
         'inference_time': cpu_time_inf,
         "flops/sample": flops}}
```

In []:

In []:

MLP Model:

```
In [18]: from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Embedding, Flatten, Dense, Dropout
from tensorflow.keras import regularizers

embedding_dim = 128

mlp_model = Sequential([
    Embedding(input_dim=vocab_size, output_dim=embedding_dim, input_length=max_len),
    Flatten(),
    Dense(256, activation='relu', kernel_regularizer=regularizers.l2(0.001)),
    Dropout(0.3),
    Dense(128, activation='relu', kernel_regularizer=regularizers.l2(0.001)),
    Dropout(0.3),
    Dense(4, activation='softmax')
])

mlp_model.compile(
    loss='categorical_crossentropy',
    optimizer='adam',
    metrics=['accuracy']
)

mlp_model.summary()
```

Model: "sequential_1"

Layer (type)	Output Shape	Param #
=====		
embedding (Embedding)	(None, 200, 128)	8320000
flatten (Flatten)	(None, 25600)	0
dense_2 (Dense)	(None, 256)	6553856
dropout_1 (Dropout)	(None, 256)	0
dense_3 (Dense)	(None, 128)	32896
dropout_2 (Dropout)	(None, 128)	0
dense_4 (Dense)	(None, 4)	516
=====		
Total params: 14,907,268		
Trainable params: 14,907,268		
Non-trainable params: 0		

```
In [19]: %%time
cpu_start_time = time.process_time()
start_time = time.time()

history_mlp = mlp_model.fit(
    X_train_pad,
    y_train_oh,
    validation_split=0.1,
    epochs=5,
    batch_size=128
)

cpu_end_time = time.process_time()
end_time = time.time()

cpu_train_time = cpu_end_time - cpu_start_time
cell_time = end_time - start_time
print(cpu_train_time)
```



```

Epoch 1/5
844/844 [=====] - 56s 65ms/step - loss: 0.5440 - accuracy:
0.8514 - val_loss: 0.3692 - val_accuracy: 0.9097
Epoch 2/5
844/844 [=====] - 53s 63ms/step - loss: 0.2738 - accuracy:
0.9470 - val_loss: 0.3782 - val_accuracy: 0.9051
Epoch 3/5
844/844 [=====] - 53s 63ms/step - loss: 0.1957 - accuracy:
0.9737 - val_loss: 0.4332 - val_accuracy: 0.8968
Epoch 4/5
844/844 [=====] - 53s 62ms/step - loss: 0.1405 - accuracy:
0.9861 - val_loss: 0.4521 - val_accuracy: 0.8913
Epoch 5/5
844/844 [=====] - 53s 63ms/step - loss: 0.1231 - accuracy:
0.9904 - val_loss: 0.4944 - val_accuracy: 0.8906
2310.71875
CPU times: total: 38min 30s
Wall time: 4min 28s

```

```

In [20]: %%time
cpu_start_time = time.process_time()
start_time = time.time()

y_pred_probs = mlp_model.predict(X_test_pad)

cpu_end_time = time.process_time()
end_time = time.time()
cpu_time_inf = cpu_end_time - cpu_start_time
cell_time_inf = end_time - start_time
print(cpu_time_inf)
print(cell_time_inf)

```

```

238/238 [=====] - 1s 6ms/step
7.25
1.4513561725616455
CPU times: total: 7.25 s
Wall time: 1.45 s

```

```

In [21]: y_pred_test = np.argmax(y_pred_probs, axis=1)

y_train_probs = cnn_model.predict(X_train_pad)
y_train_pred = np.argmax(y_train_probs,axis=1)

```

```

3750/3750 [=====] - 22s 6ms/step

```

```

In [22]: print("Training Accuracy")
train_accuracy = accuracy_score(y_train, y_train_pred)
train_f1 = f1_score(y_train, y_train_pred, average='weighted', zero_division=0)
train_report = classification_report(y_train, y_train_pred)
print('\n Training Accuracy',train_accuracy)
print('\n Training F1 Score',train_f1)
print('\n Training Classification Report',train_report)

```

Training Accuracy

Training Accuracy 0.982125

Training F1 Score 0.9821182280113712

Training Classification Report	precision	recall	f1-score	support
0	0.99	0.98	0.98	30000
1	0.99	1.00	0.99	30000
2	0.98	0.97	0.97	30000
3	0.97	0.99	0.98	30000
accuracy			0.98	120000
macro avg	0.98	0.98	0.98	120000
weighted avg	0.98	0.98	0.98	120000

```
In [23]: print("Test Accuracy")
test_accuracy = accuracy_score(y_test, y_pred_test)
test_f1 = f1_score(y_test, y_pred_test, average='weighted', zero_division=0)
test_report = classification_report(y_test,y_pred_test)
print('\n Test Accuracy',test_accuracy)
print('\n Test F1 Score',test_f1)
print('\n Test Classification Report',test_report)
```

Test Accuracy

Test Accuracy 0.8994736842105263

Test F1 Score 0.8996236037292534

Test Classification Report	precision	recall	f1-score	support
0	0.93	0.88	0.90	1900
1	0.96	0.96	0.96	1900
2	0.88	0.85	0.86	1900
3	0.84	0.90	0.87	1900
accuracy			0.90	7600
macro avg	0.90	0.90	0.90	7600
weighted avg	0.90	0.90	0.90	7600

Per each sample ,the FLOPs carried can be calculated by the formula below. This is entirely dependent on the architecture. $\text{FLOPs} = 2 \times (\text{sequence length}) \times (\text{kernel size}) \times (\text{embedding dim}) \times (\text{filters})$

```
In [24]: def analytical_flops(model):
flops = 0
for layer in model.layers:
if isinstance(layer, tf.keras.layers.Dense):
in_features = layer.input_shape[-1]
out_features = layer.output_shape[-1]
```

```

        flops += 2 * in_features * out_features
    return flops

flops = analytical_flops(mlp_model)
print(f"Analytical FLOPs: {flops} FLOPs")

```

Analytical FLOPs: 13173760 FLOPs

```

In [25]: results[('Matrix Embedding', 'MLP Model')] = {
            'Train':
                {'accuracy': train_accuracy,
                 'f1_score': train_f1,
                 'report': train_report,
                 'training_time': cpu_time_train},
            'Test':
                {'accuracy': test_accuracy,
                 'f1_score': test_f1,
                 'report': test_report,
                 'inference_time': cpu_time_inf,
                 "flops/sample": flops}}

```

In []:

In []:

In []:

```

In [26]: # create DataFrame
import pandas as pd

# flatten nested dict into a list of rows
rows = []

for (vectorizer, model), metrics in results.items():
    row = {
        "Vectorizer": vectorizer,
        "Model": model,
        "Train_Accuracy": metrics["Train"]["accuracy"],
        "Train_F1": metrics["Train"]["f1_score"],
        "Test_Accuracy": metrics["Test"]["accuracy"],
        "Test_F1": metrics["Test"]["f1_score"],
        "Inference_Time": metrics["Test"].get("inference_time", None),
        "FLOPs_per_Sample": metrics["Test"].get("flops/sample", None),
        "Training_Time": metrics['Train'].get('training_time', None)
    }
    rows.append(row)

# create DataFrame
df_results = pd.DataFrame(rows)

# optional: set a clean display order
df_results = df_results[
    ["Vectorizer", "Model", "Train_Accuracy", "Train_F1",
     "Test_Accuracy", "Test_F1", "Inference_Time", "FLOPs_per_Sample", "Training_Tim
]

```

```
print(df_results)
```

	Vectorizer	Model	Train_Accuracy	Train_F1	Test_Accuracy	\
0	Matrix Embedding	CNN Model	0.982125	0.982118	0.912105	
1	Matrix Embedding	MLP Model	0.982125	0.982118	0.899474	

	Test_F1	Inference_Time	FLOPs_per_Sample	Training_Time
0	0.912022	7.875	19660800	1685.796875
1	0.899624	7.250	13173760	1685.796875

```
In [27]: results = pd.read_csv('datasets/results.csv')
```

```
df = pd.concat([results, df_results], axis=0, ignore_index=True)
```

```
In [28]: df = df.drop('Unnamed: 0', axis=1)
df
```

Out[28]:

	Vectorizer	Model	Train_Accuracy	Train_F1	Test_Accuracy	Test_F1	Inference_Time
0	BOW	Logistic Model	0.929450	0.929328	0.902237	0.902075	0.136527
1	BOW	Random Forest Model	0.789375	0.788256	0.781579	0.779980	0.210410
2	BOW	XG Boost Model	0.930625	0.930504	0.904605	0.904392	0.184776
3	BOW	MLP Model	0.949333	0.949257	0.913026	0.912820	1.236788
4	BOW	Scratch MLP Model	0.920692	0.920560	0.902895	0.902761	3.453125
5	Tf_IDF	Logistic Model	0.919283	0.919100	0.905658	0.905387	0.302151
6	Tf_IDF	Random Forest Model	0.778833	0.779300	0.768684	0.768673	0.341379
7	Tf_IDF	XG Boost Model	0.936292	0.936206	0.903026	0.902790	0.376454
8	Tf_IDF	MLP Model	0.930417	0.930338	0.905263	0.905156	1.405875
9	Tf_IDF	Scratch MLP Model	0.896383	0.896034	0.888026	0.887549	3.312500
10	Matrix Embedding	CNN Model	0.981750	0.981750	0.913421	0.913262	8.687500
11	Matrix Embedding	MLP Model	0.981750	0.981750	0.906053	0.905959	5.921875
12	Matrix Embedding	CNN Model	0.982125	0.982118	0.912105	0.912022	7.875000
13	Matrix Embedding	MLP Model	0.982125	0.982118	0.899474	0.899624	7.250000



```
In [29]: df.to_csv('datasets/results.csv')
df_results.to_csv('datasets/matrix_emb.csv')
```

In []: